

Laminar Flow over a Circular Cylinder: Study #1

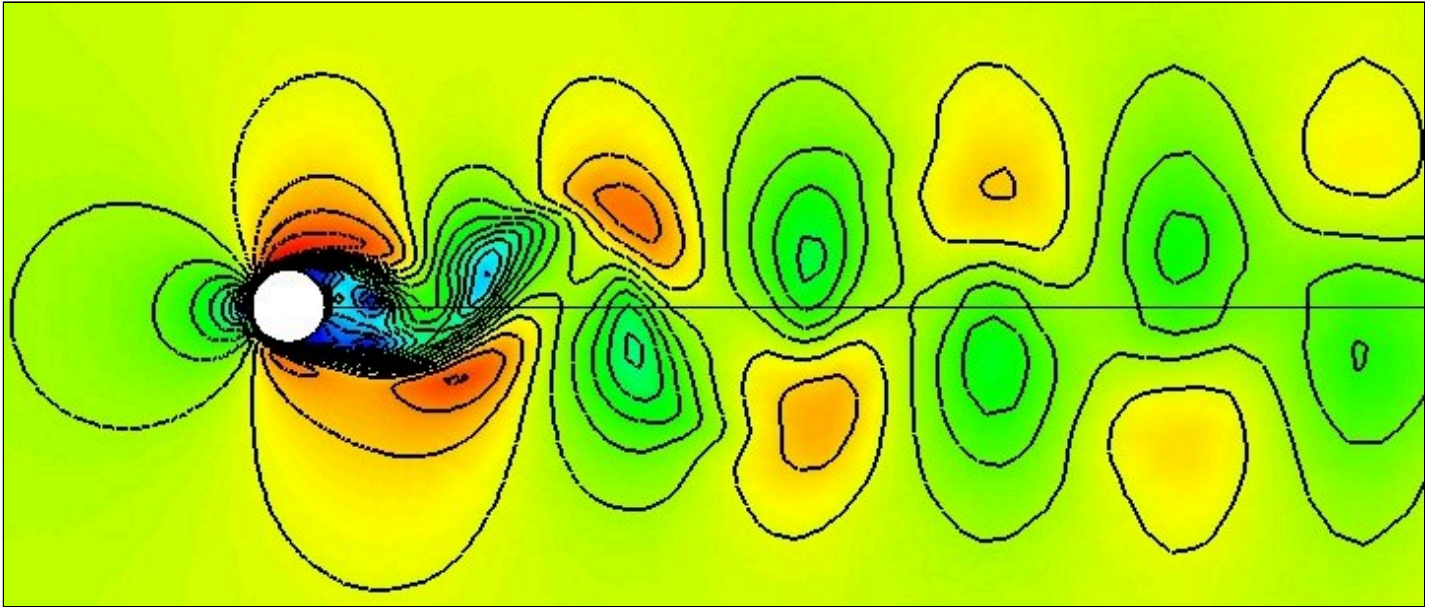


Figure 1. Mach number contours.

Introduction

This study examines laminar flow over a circular cylinder. The incoming freestream flow is uniform with a Mach number of 0.2 and a Reynolds number based on diameter of 150. At this Reynolds number, the flow is essentially two dimensional with periodic vortex pairs shed from the downstream side of the cylinder.^{1, 2} Solutions computed using the full block implicit scheme with the first and second order operators specified are compared with solutions in which the Global Newton time stepping has also been specified. The drag coefficient and Strouhal number are compared with the experimental data of Roshko.²

Download tar File

All of the archive files of this validation case are available in the Unix tar file [cyl_lam_s.tar](#). The files can then be accessed by the commands:

```
tar -xvof cyl_lam_s.tar
```

Grid

A structured C-grid was used with 401 points in the circumferential direction and 201 points in the radial direction. The grid is contained in the Wind grid file [cyl_lam_s.cgd](#). The cylinder diameter is 1.0 meter, with the J1 boundary at the cylinder wall, where the wall spacing is 1.044E-03 meters; this corresponds to a maximum y^+ of approximately 0.06. The JMAX boundary is the far-field boundary and is located 200 diameters from the center of the cylinder. The I1 and IMAX boundaries are coincident at the downstream side of the cylinder and are aligned with the freestream flow; the progression from I1 to IMAX is in the clockwise direction.

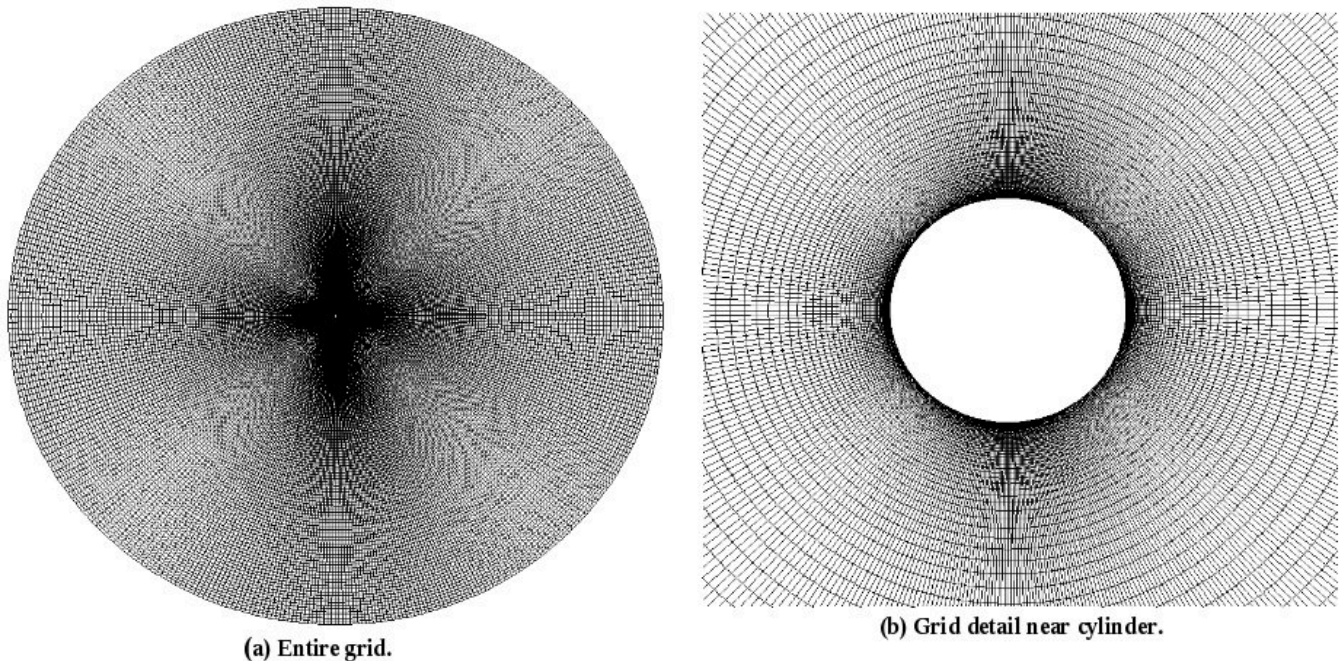


Figure 2. Computational Grids

Initial Conditions

The initial conditions are uniform flow at the freestream conditions which are given in Table 1 below. These conditions correspond to a Reynolds number of 150 based on the cylinder diameter of 1.0 meter.

Table 1. Freestream conditions.

Mach number	Static Pressure (psia)	Temperature(R)	Angle-of-Attack (deg)
0.2	0.00045146	500.0	0.0

Boundary Conditions

The boundary conditions must now be specified for the grid, and this may be done using the GMAN utility. The boundary conditions are specified as follows: a **VISCOUS WALL** boundary is specified on the J1 boundary, a **FREESTREAM** condition is specified on the JMAX boundary, and boundaries I1 and IMAX are **COUPLED** to each other.

The boundary conditions can be set either by running GMAN in an interactive mode or by creating an input data file and running GMAN in batch mode. The input file [gman.com](#) was created and gman was executed in batch mode as shown.

```
gman < gman.com
```

Computation Strategy

The computation is performed using the time-marching capabilities of Wind-US to march in a time-accurate manner to simulate the unsteady flow. For each time step, a constant global physical time step is used throughout the entire flowfield. Four individual runs are computed. The default full block implicit (left-hand-side) operator is used for all runs, with either first order (default) or second order time marching specified. The Global Newton iteration technique, aimed at stabilizing the solution and improving time accuracy, is also used for two of the runs; for each Global Newton iteration, the convergence order is 5 with 3 subiterations specified. Table 2, below, summarizes the four runs.

Table 2. Cases in terms of time marching parameters.

Run	Implicit Order	Global Newton
A	1	no
B	2	no
C	1	yes
D	2	yes

Input Parameters and Files

The input data files for the four Wind-US runs described above are named [cyl lam s.A.dat](#), [cyl lam s.B.dat](#), [cyl lam s.C.dat](#), and [cyl lam s.D.dat](#). All input parameters are identical for each run, except where noted below. The **freestream** keyword indicates that the static freestream flow conditions are specified as Mach number, pressure (psia), temperature (R), angle-of-attack (degrees), and angle-of-sideslip (degrees). The **turbulence** keyword indicates that flow is laminar. The **converge** keyword indicates that the solution will automatically stop if the maximum residual is less than 1.0e-09. The **implicit** keyword specifies the full block implicit time integration scheme to be used in all directions. The **implicit order** keyword sets the order of the implicit time marching scheme to either one or two as indicated in Table 2, above. The **implicit boundary** keyword specifies that implicit boundary conditions within the implicit solver are to be used on the cylinder wall boundary. The **rhs** keyword specifies that the second order HLLC scheme modified for structured grids is to be used. The **wall slip** keyword specifies that the velocity at the no-slip boundary is to be reduced from its initial freestream value to the no-slip condition over 1 iteration. The **cfl#** keyword specifies that the time step calculation procedure originally used in the OVERFLOW code is used; the run is specified to be time accurate and the time step in seconds nondimensionalized by L_r / a_r is 0.1.

For runs A and B, the **cycles** keyword specifies that 10000 cycles are to be computed and the **iterations** keyword specifies that there is to be 1 iteration per cycle.

For runs C and D, the **newton** keyword indicates that the Global Newton time stepping algorithm is to be used to advance 10000 Global Newton time levels with the convergence order set to 5. Within a Newton time level, the **cycles** keyword specifies 3 cycles, and the **iterations** keyword specifies 1 iteration per cycle. This effectively results in three subiterations per Global Newton iteration.

The following three keyword commands control the output. The **loads** keyword block specifies that the lift and drag coefficients on the cylinder are printed to the output list (*.lis*) file every iteration for runs A and B and at the end of every 3rd Global Newton subiteration for runs C and D. The **history** keyword block specifies that the static pressure and x-velocity at $x = 1.07$ m and $y = 0.313$ m downstream of the cylinder are written to the time history file (*.cth*) at the iteration interval specified in the **loads** keyword block. The **spawn** keyword specifies that a process is begun (or "spawned") every 20 iterations for cases A and B, and every 60 iterations for cases C and D. Note, that in cases C and D, the Global Newton subiterations are included in the iteration count, so that the time interval between spawned processes is the same for sets of runs. The process spawned is a shell script named [save cfl.sh](#), which writes the Plot3d solution file to the user created [history](#) directory (within each run directory) by calling CFPOST with [cfpost.p3d.scr.com](#). When spawning processes, it is recommended to redirect any output from the spawned process into a separate text file, in this case we named it [cfpost.out](#). Otherwise, the text generated from the spawned process gets

written to the output list (*.lis*) file, and if the RESPLT program is later used to read and postprocess this file, it erroneously reads the text from the spanwned process as residual history or loads history data; this results in corrupted RESPLT output.

Computation

The directories named [runA](#), [runB](#), [runC](#), and [runD](#) were created for each case. Within each directory, the Wind-US solver is run by entering:

```
wind -runinplace -dat cyl_lam_s.N -lis cyl_lam_s.N -flow cyl_lam_s.N -grid  
../cyl_lam_s
```

where *N* is the run identifier (**A**, **B**, **C** or **D**). Further details and options for the **wind** script can be found in the WIND-US documentation ([wind script](#)).

As Wind-US is computing each run, several output files are generated, and these files are tabulated in Table 3, below. An output list file (*.lis*) is created which contains the output from the computation including the residual information and the lift and drag coefficients for each iteration. The flow field solution files (*.cfl*) after each 20 iterations for cases A and B and 60 iterations for cases C and D are written to a directory named [history](#) within each case directory. To save disk space, only the flow files from the final solution at the last time step are included. The time history files have the *.cth* extension.

Table 3. Output file names and run times.

Run	List	Flow	Time History
A	cyl_lam_s.A.lis	cyl_lam_s.A.cfl	cyl_lam_s.A.cth
B	cyl_lam_s.B.lis	cyl_lam_s.B.cfl	cyl_lam_s.B.cth
C	cyl_lam_s.C.lis	cyl_lam_s.C.cfl	cyl_lam_s.C.cth
D	cyl_lam_s.D.lis	cyl_lam_s.D.cfl	cyl_lam_s.D.cth

Convergence

Information on the convergence history of the L2 residual can be obtained from the list files (*.lis*) using the utility RESPLT. For example, for run A, the command file [resplt.A.nsl2.com](#) contains the inputs for the RESPLT utility to output the L2 residuals to a GENPLOT formatted file [nsl2.A.gen](#) containing the residual history data.

```
resplt < resplt.A.nsl2.com
```

This process can be repeated for each case using the files corresponding to the case as follows.

```
resplt < resplt.B.nsl2.com
```

```
resplt < resplt.C.nsl2.com
```

```
resplt < resplt.D.nsl2.com
```

The program CFPOST can now be used to read in [nsl2.A.gen](#), [nsl2.B.gen](#), [nsl2.C.gen](#), and [nsl2.D.gen](#) and plot the convergence history. The CFPOST command files named [cfpost.A.l2.com](#), [cfpost.B.l2.com](#), [cfpost.C.l2.com](#), and [cfpost.D.l2.com](#), are used as follows.

cfpost < cfpost.A.l2.com

cfpost < cfpost.B.l2.com

cfpost < cfpost.C.l2.com

cfpost < cfpost.D.l2.com

The resulting plot for run A is given in Figure 3 below. As you can see, a total of 10000 iterations were computed. The residuals exhibit the flow's periodic behavior after approximately 6500 iterations. At this point, the residual has dropped four orders of magnitude. The residual plot for Run B looks very similar to Run A. For runs C and D, which use the Global Newton iteration, there are a total of 30000, because the three subiterations per overall iteration are included in the output. The L2 residual plot for Run C is given in Figure 4 and it is very similar to that for Run D.

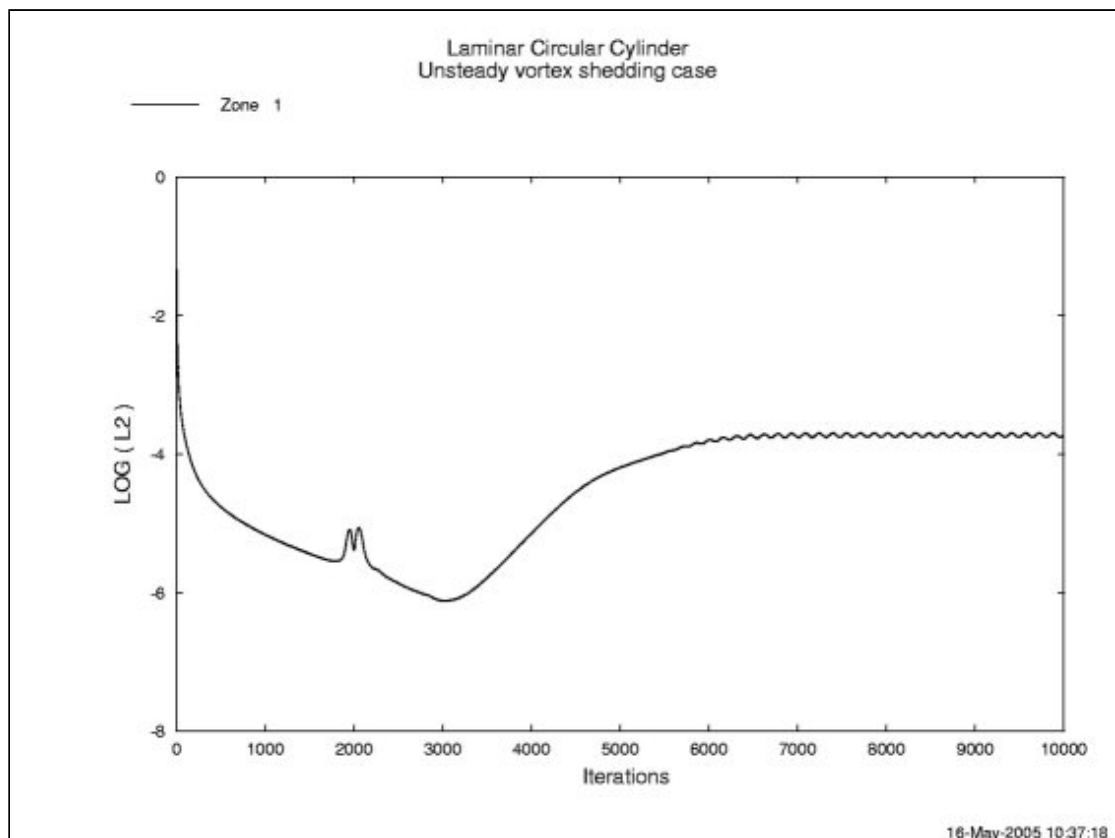


Figure 3. L2 Residual for Case A which used the first order implicit operator.

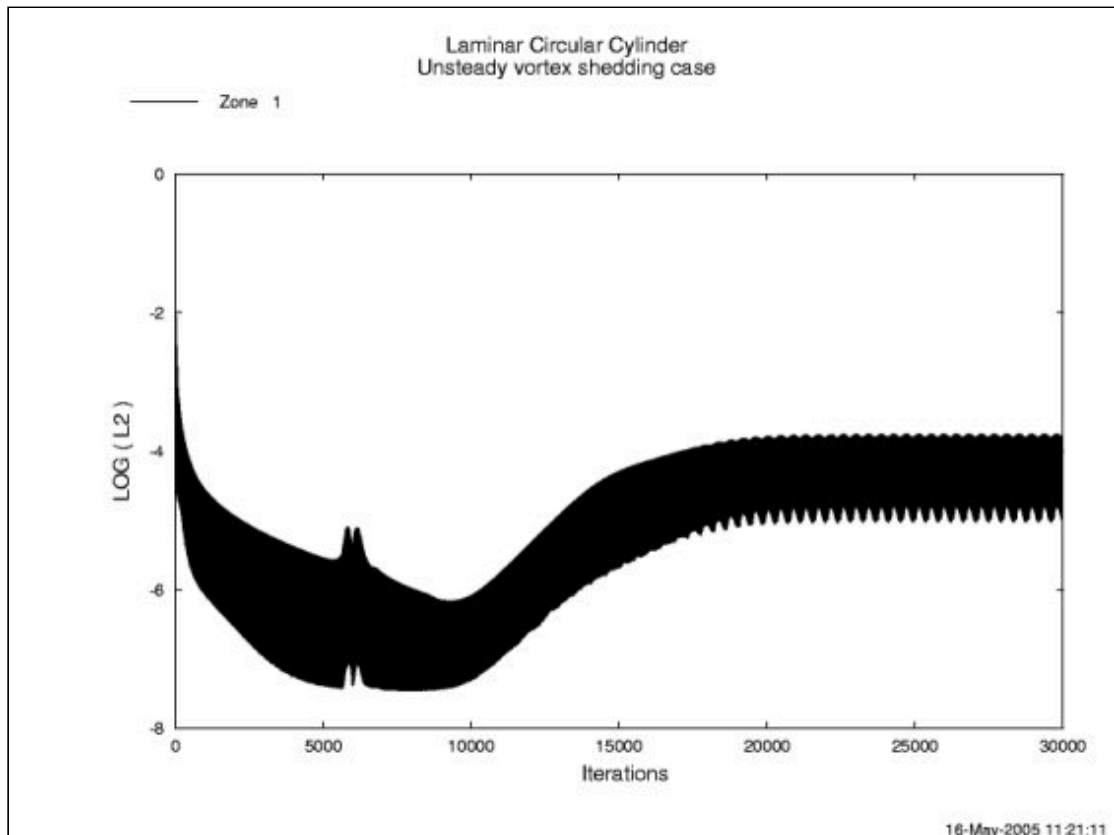


Figure 4. L2 Residual for Case C which used first order implicit and the Global Newton time stepping.

Post-Processing

One desired quantity for comparison with the literature is the average total drag coefficient. This was computed by summing the pressure drag and viscous drag coefficient values written to the .lis files as specified by the LOADS keyword block in the Wind-US input .dat files. The first step in this process is to write the lift and drag coefficients to GENPLOT formatted files.

`resplt < resplt.A.cd.com`

`resplt < resplt.B.cd.com`

`resplt < resplt.C.cd.com`

`resplt < resplt.D.cd.com`

The resulting GENPLOT formatted files are named [cd.A.gen](#), [cd.B.gen](#), [cd.C.gen](#), and [cd.D.gen](#). These files contain the pressure drag and viscous drag coefficients as well as the lift coefficient. Since we desire the total drag coefficient, which is the sum of the pressure and viscous drag coefficients, the simple Fortran90 program [cdoutput.f90](#) was written to compute this from each of the four files. The executable is called [cdoutput](#) and it is executed for each run in the manner as shown for Run A, below.

`cdoutput < cd.A.gen > cdtot.A.gen`

The resulting four files are named [cdtot.A.gen](#), [cdtot.B.gen](#), [cdtot.C.gen](#), and [cdtot.D.gen](#). The result from Run D is plotted in Figure 5 using CFPOST as follows.

`cfpost < cfpost.cdtot.D.com`

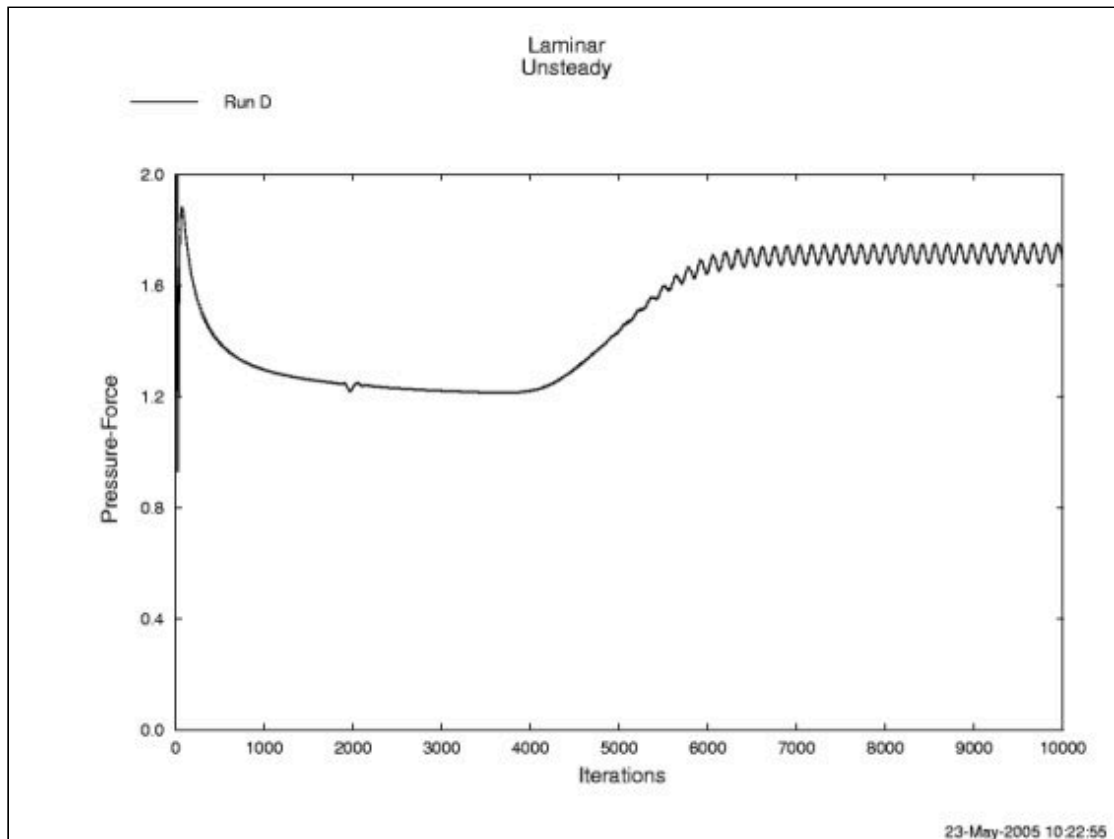


Figure 5. Total drag coefficient for Run D.

The dimensionless frequency, known as the Strouhal number, is given by $S_t = fD/U$, where D is the cylinder diameter equal to 1.0 meter and U is the freestream velocity equal to 66.8163 m/s. The parameter f is the frequency of the vortex shedding. To measure this quantity, the velocity and pressure at a point downstream of the cylinder and away from the center of the wake were written to the time history *.cth* file for each run. It's important to use a point away from the center of the wake because the center of the wake feels the influence of the vortices from both sides (Ref. 2) and will therefore give a value of the frequency double the individual vortex shedding frequency. The x -velocity is read from the time history file and written to a GENPLOT file for each nondimensional time step using the THPLT utility as shown for Run A.

thplt < thplt.A.com

The file [u.A.gen](#) results. THPLT can be run in a similar manner for Runs B, C and D, generating the files [u.B.gen](#), [u.C.gen](#) and [u.D.gen](#). The x -velocity time history for run A is shown in Figure 6.

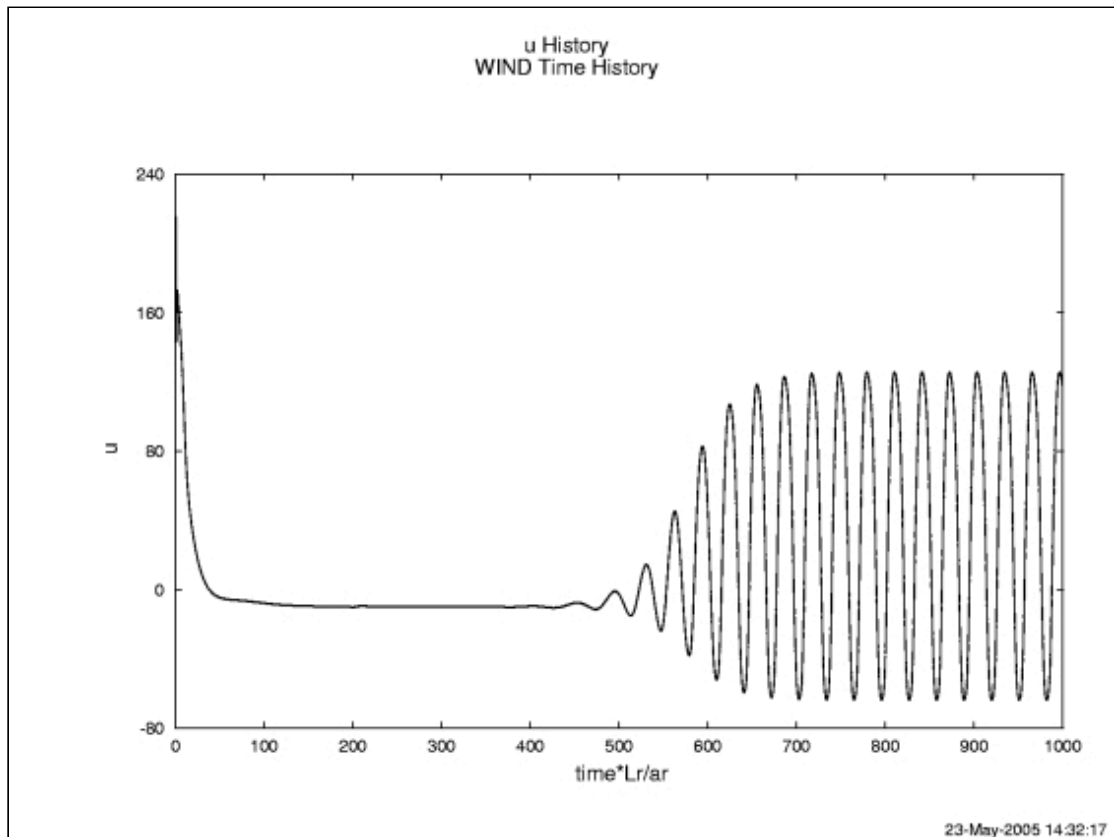


Figure 6. X-velocity history at downstream location $x = 1.07$ m and $y = 0.313$ m, Run A.

The easiest way, to compute the frequency is to use the time between peaks of the x -velocity, pressure, or lift coefficient. (The total drag coefficient should not be used to compute the frequency as it feels the influence of both vortices and will give a frequency twice that of the individual vortex shedding frequency.) For Run A, the average nondimensional time between peaks, averaged over the final four peaks, of the x -velocity is 31.0. The time in seconds is $t = 31.0 * L_r / a_r$, where $L_r = 1.0$ m and $a_r = 334.01$ m/s, and thus the frequency which is $1/t$ is equal to 10.777 1/s. The Strouhal number is then computed using the above formula to be 0.161. When the frequency is computed in this manner, the Strouhal number will be referred to as S_{t1} . An alternate way of computing the frequency is to use a Fast Fourier Transform (FFT),⁴ of the velocity data over a given number of iterations. One may write a program to do this, or use mathematical software such as MATLAB⁵. When using MATLAB to compute the frequency, using the last 2000 points and the last 3000 points, the respective Strouhal numbers, S_{t2} and S_{t3} , are 0.150 and 0.1667 for Run A.

Comparisons of the Results

In Figure 6, the total drag coefficient is plotted for all four runs using CFPOST.

cfpost < cfpost.cdtotall.com

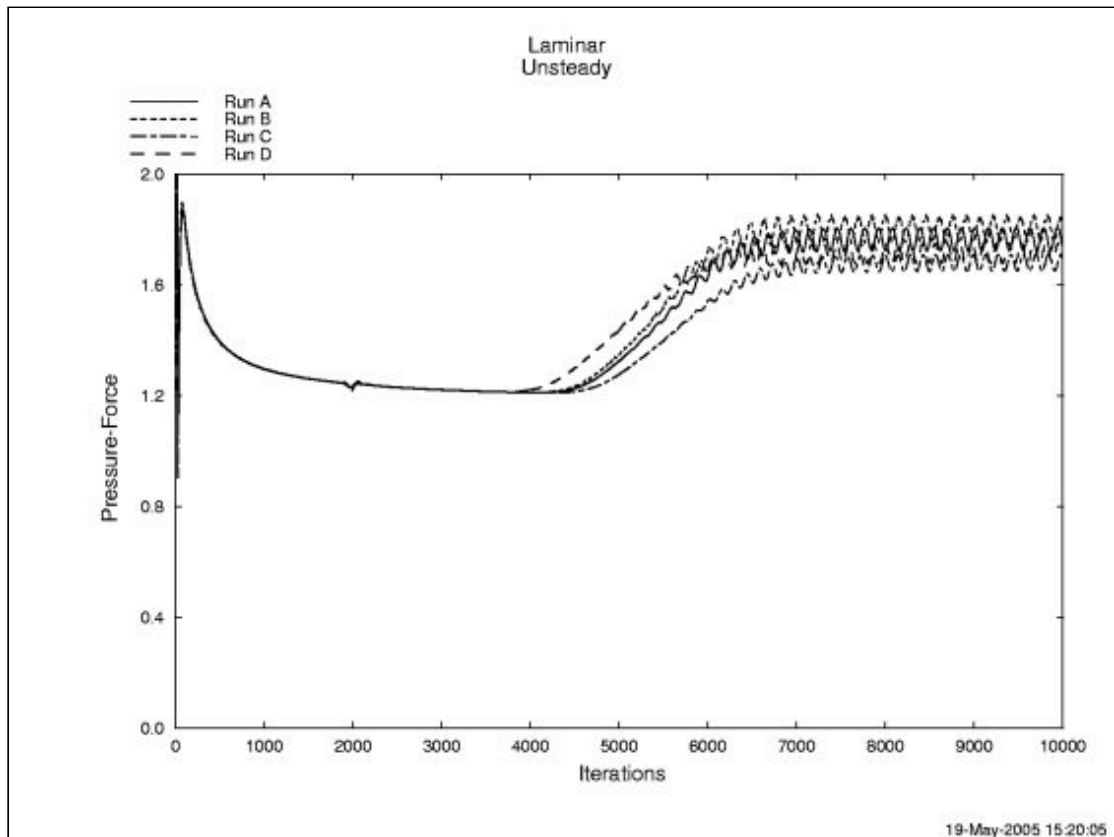


Figure 7. Total drag coefficient for all four runs.

The program [cdoutput](#), used above to compute the total drag coefficient, also computes the average drag coefficient over the last 8000 iterations. This value is listed in the last line of each GENPLOT output file above and is also tabulated in Table 4. The experimental value is 1.78.³ Table 4 also gives the Strouhal numbers computed using the three methods described in the previous section where S_{t1} is the value obtained by hand-calculations of the frequency using the average time between the last four peaks, S_{t2} is the value obtained using an FFT over the last 2000 iterations to compute the frequency, and S_{t3} is the value obtained using an FFT over the last 3000 iterations. The experimental value of the Strouhal number ranges from 0.179 - 0.182.¹

Table 4. Average Drag Coefficient and Strouhal Number.

Run	Avg C_D	S_{t1}	S_{t2}	S_{t3}
A	1.767	0.161	0.150	0.167
B	1.804	0.164	0.175	0.167
C	1.679	0.177	0.175	0.183
D	1.715	0.180	0.175	0.183

Sensitivity Studies

As described above, this study compared the use of the first order versus the second order operator in the block implicit method, with and without the Global Newton iteration scheme.

Performance

Wind-US version 1.123 was run on an HP xw6200 workstation running the LINUX operating system. The cpu time per iteration and the overall time to compute each case, as reported in the output list files, is given in Table 5.

Table 5. Run times.

Run	CPU time(iter-node)	Total CPU time
A	16.93 microsec	13646 sec
B	16.97 microsec	13678 sec
C	12.99 microsec	31415 sec
D	17.23 microsec	41663 sec

Acknowledgements

These cases were originally run by Chris Nelson, Senior Scientist, Innovative Technology Applications Company, LLC, who provided all of his grid, input and output files, as well as valuable insights, so that this validation case could be composed. The cases were re-run, with minor modifications, by Julianne C. Dudek, at NASA Glenn, who composed this web page.

References

1. Nichols, R. H., and Heikkinen, B. D., "Validation of Implicit Algorithms for Unsteady Flows Including Moving and Deforming Grids," AIAA Paper 2005-0683, 2005.
2. Roshko, A., "On the Development of Turbulent Wakes from Vortex Streets," NACA Report 1191, 1954.
3. Schlichting, H., Boundary-Layer Theory, Sixth Edition, p 17, McGraw-Hill Book Company, 1968.
4. Numerical Recipes in FORTRAN 77: The Art of Scientific Computing,, (ISBN 0-521-43064-X), Copyright 1986-1992 by Cambridge University Press.
5. MATLAB, Copyright 1984-2002, Mathworks, Inc.

Contact Information

This case was created on June 6, 2005 by [Chris Nelson](#), who may be contacted at:

Innovative Technology Applications Company, LLC
Phone: (425) 778-7853
e-mail: [Chris Nelson](#)

and [Julianne C. Dudek](#), who may be contacted at:

NASA John H. Glenn Research Center, MS 86-7
21000 Brookpark Road
Brook Park, Ohio 44135
Phone: (216) 433-8513
e-mail: Julianne.C.Dudek@nasa.gov

Responsible NASA Official/Curator: [John W. Slater](#)

Web Policies: [Web Privacy Policy and Important Notices](#)

Adobe Reader Download: [Adobe Reader](#)